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ORIGINAL RESEARCH ARTICLE

Factors Associated with Persisting Post-Concussion Symptoms Among Collegiate Athletes and Military Cadets: Findings from the NCAA-DoD CARE Consortium

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Abstract

Background Persisting post-concussion symptoms (PPCS) is a condition characterized by prolonged recovery from a mild traumatic brain injury (mTBI) and compromised quality of life. Previous literature, on the basis of small sample sizes, concludes that there are several risk factors for the development of PPCS.

Objective We seek to identify protective and risk factors for developing slow recovery or persisting post-concussion symptoms (PPCS) by analyzing medical history, contact sport level, setting, and the Sport Concussion Assessment Tool (SCAT) and Brief Symptom Inventory (BSI-18) assessments at baseline and post-injury.

Patients and Methods We studied 47,860 unique collegiate athletes and Military Service Academy (MSA) cadets enrolled in the 30-site National Collegiate Athletic Association and Department of Defense (NCAA-DoD) Concussion Assessment Research and Education (CARE) Consortium prospective cohort study of baseline and post-injury data from 2014 to 2019. Medical histories and preseason baselines ($n = 60{,}720$), along with SCAT and BSI-18 examinations ($n = 5379$) conducted within 48 h post-injury, were analyzed. PPCS is defined as cleared for return to play (RTP) > 29 days.

Results Of 5073 concussions, the median (95% CI) and mean (SD) RTP were 15.2 (4.8–82.6) days and 22.2 (23.0) days, respectively, with 891 developing PPCS. Sex, high SCAT score, high BSI-18 score, and delayed reporting produced small effect sizes on RTP ($d = 0.22$–0.44). Adjusted odds ratios (OR) of developing PPCS indicated the following risk factors: SCAT total score > 45 (OR = 1.91, 95% CI: 1.58–2.30), female sex (OR = 1.80, 95% CI: 1.53–2.13), concussion history (OR = 1.80, 95% CI: 1.29–2.52), and delayed reporting (OR = 1.42, 95% CI: 1.20–1.67). In contrast, protective factors against developing PPCS were: being a limited-contact (OR = 0.34, 95% CI: 0.25–0.47), noncontact (OR = 0.35, 95% CI: 0.24–0.51) or contact sport athlete (OR = 0.34, 95% CI: 0.28–0.41); and receiving the concussion at practice (OR = 0.64, 95% CI: 0.53–0.77) or in competition (OR = 0.44, 95% CI: 0.34–0.56). Athletes diagnosed with a learning disability and taking a neurostimulant were more protected than those who were not (OR = 0.44, 95% CI: 0.26–0.76 versus OR = 0.72, 95% CI: 0.49–1.05, respectively). A prognostic model using these variables offers poor sensitivity (9%) but high specificity (98%) in identifying PPCS (AUC = 0.72).

Conclusions This study of 5073 concussions indicates that female sex, high symptom burden, and prior concussion are risk factors for slow recovery. In addition, learning disability, contact sport, and concussion in competition are protective against slow recovery. The neurostimulant results suggest that clinicians should keep their attention deficit disorder (ADD) and attention-deficit/hyperactivity disorder (ADHD) patients on their current neurostimulant medication after a concussion. A prognostic logistic regression model based on behavioral clinical findings did a poor job of identifying PPCS.

Franco Pestilli and Nicholas L. Port shared senior contribution.

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Key Points

Findings: Using 5073 concussed athletes, this case–control study finds that type of sport, injury setting, and learning disability are protective for persisting post-concussion symptoms, whereas sex, concussion history, high symptom burden, and delayed reporting are risk factors for PPCS.

Meaning: For the first time, this study provides evidence of factors that are protective against slow concussion recovery and suggests patients on neurostimulants should remain on their medication.

1 Introduction

A concussion is a change in brain function following an accelerative blow to the head or body, sometimes accompanied by temporary loss of consciousness (LOC). It is identified in awake individuals using measures of neurologic and cognitive dysfunction [1]. Symptoms can arise acutely or the following day and differ in their profile, severity, and time to resolution. One of the most serious post-concussion concerns is persisting post-concussion symptoms (PPCS or post-concussion syndrome). Generally agreed upon as symptoms lasting longer than 1 month post mild traumatic brain injury (mTBI) or concussion [2], PPCS involves debilitated quality of life and problems returning to school or work. Identifying protective and risk factors is clinically important [3, 4], and both preexisting and post-injury factors have been proposed.

Preexisting factors thought to increase PPCS risk include mental health history (especially anxiety and depression), learning disability (i.e., dyslexia or attention deficit disorder [ADD]), migraine history, prior concussion, female sex, helmeted sport, genetics, and age [4, 5]. Anxiety, anxiety disorder, and depression are all common mental health conditions [6]. In a study of 67 athletes, one-third reported anxiety symptoms post-concussion, though those with baseline symptoms (but no diagnosed anxiety disorder) experienced no increase in depression or anxiety symptoms post-concussion. Among 247 collegiate athletes, those with pre-injury anxiety took twice as long to recover as those without pre-injury anxiety, and those with diagnosed depression took 2.2 times longer to recover than those without depression [7]. Those with ADD report more symptoms on the Sport Concussion Assessment Tool (SCAT3) symptom scale at baseline [8], but Broglio and colleagues [3] found a protective effect for return to play (RTP) time when student athletes with ADHD were medicated. Migraine history is an important factor in concussion management, and a study of 3063 children with acute concussions showed an association between migraine history and longer-lasting symptoms [9]. The inclusion of "headache" in concussion assessment tools complicates the exploration of migraine history impact, as they can be indicative of migraine disorder [5]. Several studies suggest that the number of prior concussions is a risk factor for PPCS. Guskiewicz and colleagues [10] found athletes ($N = 2905$) with three or more had longer-lasting symptoms than those with one. Slobounov and colleagues [11] showed that athletes ($N = 160$) with no prior history were slower to recover from a second concussion than the first. Among 247 children and adolescents, those with three or more prior concussions took 3.6 times longer to return to a normal school schedule than students with no prior history [7]. Female sex has been found to increase the risk of concussion symptoms by about 28% [12], but PPCS risk has not been explored. Several relatively small (N values from 260 to 294) studies have measured the role of contact sport level in PPCS risk. Two found helmeted sports were protective for PPCS [4, 13] and one found that non-helmeted sports increased risk [4].

Post-injury factors thought to increase PPCS risk include LOC [14], higher post-injury test scores (Brief Symptom Inventory (BSI-18) and SCAT symptoms), injury setting (practice versus competition play), and delayed reporting [15]. In a sample of 247 athletes, Corwin and colleagues [7] found that those with LOC at injury took 1.8 times longer to recover. The BSI-18 is an 18-question mental health survey assessing depression, anxiety, and somatization symptoms. Among 2055 athletes, higher BSI-18 somatization was linked to delayed recovery [16]. The SCAT is used in concussion management and includes symptom survey and cognitive, balance, physical, and neurological assessments [17]. Several studies find an association between high symptom score and slower recovery, putting athletes at risk for PPCS [18–21]. Few studies have examined injury setting (practice, competition, or outside of sport) and PPCS, though Broglio and colleagues (using Concussion Assessment Research and Education (CARE) data) found those with practice-based injuries took 2 days longer to initiate a RTP protocol compared with those with competition-based injuries [3], and Asken and colleagues found (also using CARE) that delayed symptom reporting increased risk for longer recovery [22, 23]. Bookbinder and colleagues [24] found females were more likely to delay reporting.

Thus, while preexisting and post-injury factors are hypothesized to increase PPCS risk, there are few definitive answers. Studies are scarce, sample sizes tend to result in underpowered hypothesis testing, and many are limited to either pre- or post-injury factors. A multiple-variable approach is ideal, as any given athlete will have several potentially relevant pre- and post-injury factors. Multiple-variable approaches have been published several times in the sports-related concussion (SRC) literature with the number of subjects ranging from 40 to 655 [25–28].

The present study addresses these shortcomings using the CARE dataset, which includes 60,720 baseline examinations and 5379 acute post-injury examinations (< 48 h). All participants completed extensive medical histories and were followed post-injury via standardized protocol. Our goal was to build a clinically useful prognostic model of PPCS.

2 Patients and Methods

2.1 Participants

Analyses included 47,860 unique collegiate athletes and Military Service Academy (MSA) cadets from the 30-site, multicenter, National Collegiate Athletic Association and Department of Defense (NCAA-DoD) CARE Consortium study [29] from 2014 to 2019, which includes sites from all three NCAA divisions. There were 60,720 preseason baseline examinations, including 36,608 single, 9650 double, 1596 triple, and 6 quadruple baselines. The mean (SD) age of the participants was 19.2 (± 1.4) years. At post-injury (< 48 h), 5379 examinations were analyzed from 4726 athletes, including 4158 single, 497 double, 57 triple, and 14 quadruple concussions (Fig. 1). There are missing data, which changes the number of subjects throughout this study, e.g., although all data collection sites were supposed to conduct the BSI-18 survey, they often did not. Since all participants in CARE were active athletes who were cleared to participate in collegiate sports, we did not exclude any subjects from our analysis because of medical history. Not surprisingly, in this study of 5073 concussions, there were participants who self-reported a history of a neurological condition, including a history of meningitis ($n = 33$), seizure disorder ($n = 24$), sleep disorder ($n = 74$), balance disorder ($n = 43$), moderate/severe TBI ($n = 33$), brain surgery ($n = 6$), and stroke ($n = 6$). A total of 553 nonsport cadets were excluded from the analysis. A nonsport cadet is a cadet who did not participate in an NCAA-sanctioned sport. The CARE study was a convenience sample to examine the natural history of concussion, so no power analysis was performed.

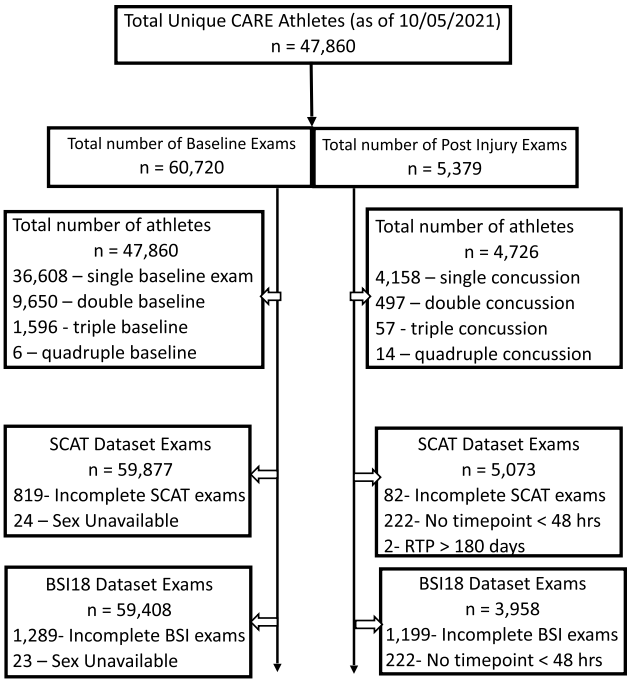

Fig. 1 CONSORT diagram for inclusion and exclusion criteria. Concussion Assessment Research and Education (CARE), number (*n*), Sport Concussion Assessment Tool (SCAT), Brief Symptom Inventory (BSI-18), hours (hrs), Return to Play (RTP)

DatalysCenter managed the clinical research core data of the CARE Consortium. This study was approved by the human research protections office and the institutional review boards at each participating institution. All subjects provided written consent.

2.2 Baseline Examination Exclusions

Baseline examinations ($n = 60{,}720$) were separated into two datasets on the basis of either BSI-18 or SCAT3 examination completion status. In the BSI-18 baseline dataset, 1289 incomplete BSI-18 examinations and 23 athletes with sex unavailable left 59,408 baseline examinations. In the SCAT3 baseline dataset, 819 incomplete SCAT3 examinations and 24 athletes with sex unavailable left 59,877 baseline examinations. The BSI-18 baseline dataset was only used to examine BSI-18 total score.

2.3 Post-Injury Examination Exclusions

Post-injury examinations ($n = 5379$) were also separated into two datasets based on BSI-18 or SCAT3. In the BSI-18 dataset, 1199 incomplete BSI-18 examinations and 222 with a time point greater than 48 h left 3958 examinations. In the SCAT3 dataset, 82 incomplete SCAT3 examinations and 222 with a time point of greater than 48 h left 5075 examinations. Two subjects were excluded due to RTP > 180 days, leaving 5073 examinations. The BSI-18 post-injury examination dataset was only used to examine BSI-18 total score.

2.4 Data Collection

2.4.1 BSI-18 Examination

The BSI-18 is an 18-question mental health survey of depression, anxiety, and somatization symptoms with good test–retest reliability and internal reliability [12]. Each subscale includes six rating scale (0 = "not at all" to 4 = "extremely") questions (total scores 0–72). For both BSI-18 and SCAT3, we chose to stratify athletes with high symptom scores by using an 80th–20th percentile threshold. The top 20th percentile is an arbitrary value that captures those who are very symptomatic. For BSI-18, a score of 10 or greater places a subject in the top 20th percentile of symptom scores.

2.4.2 SCAT3 Examination

The SCAT3 is used in concussion management and includes symptom ratings and cognitive, balance, physical, and neurological assessments [17]. The current edition, SCAT5, was developed during CARE data collection and not used in CARE. This analysis uses only the symptom evaluation, which is identical in SCAT3 and SCAT5 [30]. The symptom evaluation consists of 22 self-report symptoms on a 7-point rating scale (0 = "none" to 6 = "severe", total range 0–132). A score of 45 or greater places an athlete in the top 20th percentile of symptom scores.

2.4.3 Medical History

Self-reported medical histories on a standardized intake form included learning disability, mental health, prior concussion, migraine, LOC, type of sport (in terms of contact), injury setting, current medications, and delayed reporting. "Yes" to the question: "Have you ever been diagnosed by a Physician/Doctor of Medicine (MD) with Depression," or the same question with "Anxiety Disorder," indicated a positive mental health history. A positive learning disability history was assessed via the question: "Have you ever been diagnosed by a Physician/MD with Learning Disorder" or "ADD/ADHD." We separately assessed ADHD medication use via the question, "Are you currently taking a psychostimulant (e.g. ADD/ADHD medications)" at baseline and post-injury. For athletes indicating a learning disability or ADD/ADHD, we also examined every self-reported medication at baseline and post-injury. Athletes taking Adderall, Ritalin, methylphenidate, Concerta, Vyvanse, amphetamine, Focalin, or other ADD/ADHD medications were identified as taking a psychostimulant. In this search, approximately 64 spelling variants were observed. We presumed that athletes would stay on that medication throughout college. Athletes with a self-reported history of three or more concussions prior to their baseline examination were identified as having a positive concussion history. Concussions accrued during CARE data collection were added to their baseline self-reported number. Three or more was chosen as the cutoff on the basis of previous literature. A positive migraine history was indicated by an athlete answering "yes" to "Have you ever been diagnosed by a Physician/MD with migraine headaches." LOC was denoted by loss of consciousness at the time of injury. Injury setting had three categories: practice, competition, and outside of sport. Delayed reporting was defined as reporting a concussion one or more days post-injury.

2.4.4 Sport Category

Sports were separated into four categories: contact (football, diving, basketball, lacrosse, soccer, water polo, wrestling, ice hockey, and field hockey); limited contact (volleyball, track and field, cross country, softball, baseball, gymnastics, beach volleyball, and fencing); noncontact (swimming, tennis, rowing, cheer, golf, sailing, rifle, and bowling); and cadet (many different sports). The CARE Consortium sport categories are based on a paper by Rice and colleagues [31].

2.4.5 Return to Play Time

The RTP decision was made by sports medicine staff at each institution. All institutions followed the graded return protocol of the 5th International Conference on Concussion in Sport [32]. The medical staff performed a comprehensive examination beyond what was required and recorded by the CARE Consortium.

2.5 Analytical Design

At baseline and post-injury, the dependent variables examined were SCAT symptom total score and BSI-18 total score. Independent variables were mental health history, learning disability (with and without neurostimulant), prior concussion, migraine history, LOC, sex, sport category, injury setting, and delayed reporting. Clinically defined

RTP and binary PPCS categorization (RTP > 29 days) were additional dependent variables, with the same independent variables.

2.6 Statistical Methods

The descriptive mean, standard deviation (SD), median, and 95% confidence interval (CI) at baseline and post-injury were calculated for all variables of interest, with SD and 95% CI rounded to the nearest integer. Kolmogorov–Smirnov significance and Cohen's *d* effect size were calculated at baseline and post-injury. Pearson correlation coefficients between continuous variables and odds ratios for binary variables [33] were calculated. Null hypothesis testing on RTP survival functions was performed with the Kolmogorov–Smirnov test for bivariate data and with the Kruskal–Wallis test for multivariate data. Cox proportional hazard models were performed for the association of variables on the RTP survival functions, yielding a hazard ratio for each variable. In this modeling, when there are multiple categories, one group serves as the reference against which the other groups are compared. Odds ratios [34] were calculated for the independent variables. Adjusted odds ratios were calculated via a variable reduction stepwise logistic regression (LR) in R (stepAIC); overfitting and accuracy were checked by LASSO regression [35, 36]. Traditionally the order of variables could affect the results of stepwise variable reduction, thus stepAIC in R was run with the forward, backward, and bidirectional parameter selected, and the model produced an identical result each time. The same LR was used to estimate PPCS as a function of history, sex, and baseline and post-injury examination data. We removed two subjects with an RTP time beyond the 99.9% confidence interval cutoff (both greater than 309 days). Missing data, which are inevitable in large multicenter clinical studies, are documented in every panel of the analysis with the number of subjects listed. In the multivariate analyses, missing data were coded as not a number and ignored. The analysis code (95% written in MATLAB, 5% written in R [37]) is publicly available at: https://github.com/Nicholas-Port/Persistent-Post-Concussion-Symptoms. Much of the CARE dataset is available at: https://fitbir.nih.gov/.

3 Results

3.1 Association of Preexisting Conditions on BSI-18 and SCAT Symptom Baseline Questionnaires

Overall, the association of preexisting conditions on BSI-18 total scores was unremarkable. Mental health history had a large effect size ($d = 1.01$), while learning disability ($d = 0.18$), prior concussion ≥ 3 ($d = 0.23$), migraine history ($d = 0.26$), and sex ($d = 0.25$) had small effect sizes. The overall association of preexisting conditions on SCAT symptom scores was also unremarkable. Mental health history had a medium effect size ($d = 0.56$), while learning disability ($d = 0.11$), prior concussion ($d = 0.21$), migraine history ($d = 0.22$), and sex ($d = 0.18$) had small effect sizes (Supplementary Fig. 1). Owing to missing concussion history, 793 athletes were excluded.

3.2 Association of Preexisting Conditions on BSI-18 and SCAT Symptom Acute Post-Injury Questionnaires

The mean time between injury and acute examination was 1.8 days with a SD of 2.4 days. At the acute post-injury time point, the association of preexisting conditions on BSI-18 scores was indifferent. Mental health history ($d = 0.43$) had a medium effect size, whereas prior concussion history (≥ 3 concussions, $d = -0.09$), migraine history ($d = 0.12$), LOC ($d = 0.07$), sex ($d = 0.18$), and delayed reporting ($d = 0.22$) had small effect sizes. The association of preexisting conditions on SCAT symptom scores was also indifferent. Mental health history ($d = 0.33$) and delayed reporting (0.33) had medium effect sizes, whereas prior concussion ($d = -0.15$), migraine history ($d = 0.13$), and sex ($d = 0.25$) had small effect sizes (Supplementary Fig. 2). Owing to missing prior concussion history, 96 athletes were excluded. Effect sizes of preexisting conditions on BSI-18 and SCAT at baseline and acute post-injury were sufficiently small as to not be incorporated into subsequent analyses.

3.3 Association of Preexisting Conditions on RTP

The median and mean RTP were 15.2 and 22.2 days, respectively, with a range of 2–308 days. Female sex, high BSI-18 score, and delayed reporting had small effect sizes (0.25, 0.26, and 0.22, respectively) (Fig. 2). Mental health, prior concussion, migraine history, and LOC had non significant effect sizes ($d = 0.06$, $d = 0.06$, 0, and 0.07, respectively). High SCAT score had a medium effect size (0.36). Conversely, learning disability, NCAA sport category (contact, limited contact, noncontact, cadet), and injury setting (competition, practice, and outside sport) had significant effect sizes on RTP (Kruskal–Wallis ANOVA, $p < 0.001$). Note that 698 athletes had a missing RTP time and 72 athletes had a missing concussion history (≥ 3) and were excluded from analysis. Learning disability, NCAA category, and injury setting are nonbinary, multiple categorical comparisons and require a reference variable be chosen in the Cox proportional hazards regression model (Supplementary Table 1). In these analyses, the rightmost distribution for each variable is the reference group (e.g.,

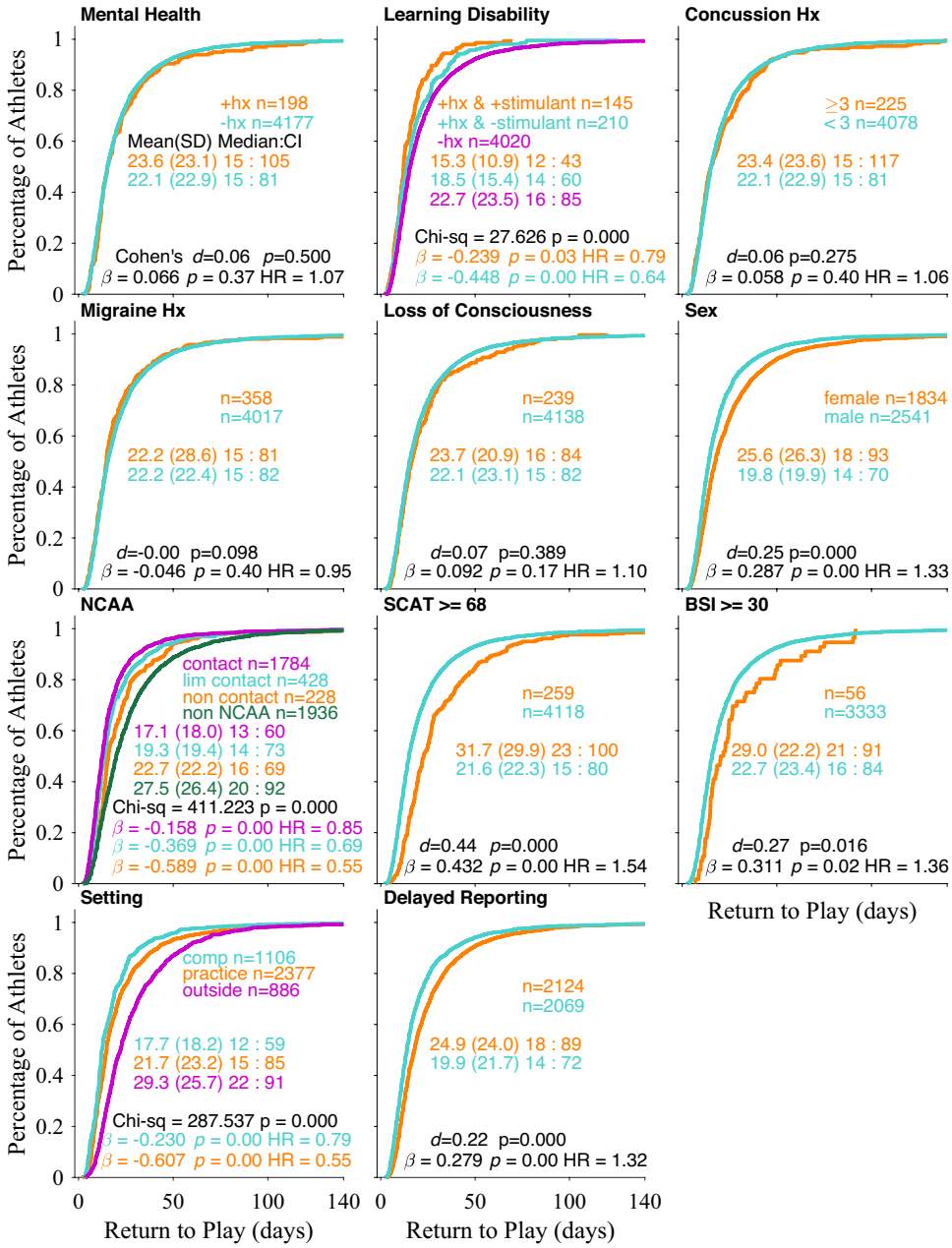

Fig. 2 Effect of preexisting conditions on days to RTP. Cumulative histograms of RTP time for those with and without preexisting conditions of mental health history, learning disability (with and without neurostimulant), prior concussion, migraine history, loss of consciousness, female sex, type of contact sport, high SCAT score, high BSI-18 score, setting, and delayed reporting. Orange histograms indicate those with a positive history and cyan histograms indicate those with a negative history. The NCAA category is divided into four groups: contact, limited contact, noncontact sports, and cadet. The number of athletes (n) is provided on each graph. The descriptive statistics are mean (standard deviation) and median: 95% confidence interval. Cohen's d effect size and Kolmogorov–Smirnov nonparametric null hypothesis testing probability are reported for every panel except NCAA category. For NCAA category, Kruskal–Wallis ANOVA (chi-squared and p value) and a Cox proportional hazard model (beta weight of the covariate and p-value) are reported. Hazard ratios (HR) are provided. History (Hx), National Collegiate Athletics Association Sport Category (NCAA), Sport Concussion Assessment Tool (SCAT), Brief Symptom Inventory (BSI-18), Competition (comp)

learning disability negative history, cadet, and outside of sport concussion). A breakdown of mechanism of injury for the NCAA category of cadet is available in Supplementary Fig. 3.

3.4 Odds Ratios of Developing PPCS

Odds ratios are used to explain either a protective (0 to < 1) or risk (> 1) association for each factor. PPCS is defined as RTP > 29 days. Risk factor odds ratios (Table 1) were: mental health (OR = 1.16), BSI-18 total score > 10 (OR = 1.60), SCAT symptom score > 45 (OR = 1.98), prior concussion (OR = 1.29), LOC (OR = 1.18), sex (OR = 1.86), cadet (OR = 3.04), concussion outside of sport (OR = 2.62), and delayed reporting (OR = 1.81). Protective factor odds ratios were learning disability (OR = 0.59), learning disability with neurostimulant (OR = 0.44), learning disability without neurostimulant (OR = 0.72), migraine history (OR = 0.93), noncontact sport (OR = 0.93), limited contact sport (OR = 0.68), contact sport (OR = 0.34), concussion in practice (OR = 0.80), and concussion in competition (OR = 0.48). The reciprocal is reported for all odds ratios as a convenience.

Adjusted odds ratios were produced using a stepwise LR model (Fig. 3A). Bidirectional stepwise variable

reduction was completed prior to running the model to discard factors with no meaningful PPCS association. Variables removed included mental health history, high BSI-18 score, learning disability, migraine history, LOC, cadet, concussion outside of sport, and learning disability with and without a neurostimulant. Adjusted odds ratios of risk factors for PPCS were SCAT symptom score > 45 (OR = 1.91), prior concussion history (≥ 3) (OR = 1.80), female sex (OR = 1.80), and delayed reporting (OR = 1.42). Protective factors were being a noncontact sport athlete (OR = 0.35), limited contact sport athlete (OR = 0.34), contact sport athlete (OR = 0.34), sustaining a concussion at practice (OR = 0.64), and sustaining a concussion in competition (OR = 0.44). Adjusted odds ratios were calculated with multiple-binary-variable LR. For the multiple-categorical variables (learning disability, NCAA category, and injury setting), the last category was the reference and had no adjusted odds ratio (learning disability history, cadet, and outside of sport concussion) [38]. The odds ratios when the cadets were removed from the dataset were noncontact (OR = 1.79), limited contact (OR = 1.32), and contact (OR = 0.62). The raw two-by-two tables are shown in Supplementary Figs. 4 and 5 (PPCS > 29 days).

3.5 Correlation Between Questionnaires and Days to RTP

Ideally, clinical examination variables should have a strong association with clinical outcomes. For athletes with a negative history for mental health, learning disability, prior concussion, migraine, LOC, or delayed report, there were weak correlations ($r < 0.24$) between RTP and either BSI-18 total score or SCAT total symptom score. For student athletes with a positive history for mental health, learning disability, prior concussion, migraine, or LOC, and delayed report, only mental health ($r = 0.21$) and learning disability ($r = 0.23$) had weak, although significant, correlations between BSI-18 total score and RTP. Mental health ($r = 0.22$) and concussion history ($r = 0.23$) had weak, although significant, correlations between SCAT total symptom score and RTP. The scatter plots are illustrated in Supplementary Fig. 6.

3.6 Logistic Regression Model of PPCS

Model inputs were all variables produced from the previous stepwise regression (prior concussion, high SCAT, female, delayed reporting, practice, competition, contact, noncontact, and limited contact; Fig. 3A). The model had an area under the curve (AUC) of 0.72, a low sensitivity of 9%, and a high specificity of 98% (Fig. 3B). To check for overfitting and promote sparsity in our logistic regression model, a LASSO regression model was also performed and produced nearly identical results in Supplementary Table 2.

Table 1 Unadjusted odds ratios between factors and RTP > 29 days, adjusted odds ratios using an AIC stepwise regression model between factors and RTP > 29 days, and confidence intervals (CI) for the adjusted odds ratios

History	Odds ratio	95% CI	Adj. odds ratio	95% CI
Mental health	1.16	0.82–1.63	–	–
High BSI-18	1.60	1.32–1.95**	–	–
High SCAT	1.98	1.67–2.34**	1.91	1.58–2.30**
Learning disability	0.59	0.43–0.81*	–	—
Prior concussion	1.29	0.95–1.77*	1.80	1.29–2.52**
Migraine	0.93	0.71–1.22	—	—
Loss of consciousness	1.18	0.87–1.61	—	—
Female sex	1.86	1.60–2.16**	1.80	1.53–2.13**
Noncontact	0.93	0.67–1.31	0.35	0.24–0.51**
Limited contact	0.68	0.51–0.89	0.34	0.25–0.47**
Contact	0.34	0.28–0.40**	0.34	0.28–0.41**
Cadet	3.04	2.61–3.55**	–	–
Practice	0.80	0.69–0.93*	0.64	0.53–0.77**
Competition	0.48	0.40–0.59**	0.44	0.34–0.56**
Outside	2.62	2.23–3.09**	–	–
Delayed reporting	1.81	1.55–2.10**	1.42	1.20–1.67**
+ Stimulant	0.44	0.26–0.76*	–	–
− Stimulant	0.72	0.49–1.05	–	–

*Denotes p-value < 0.01, ** denotes p-value < 0.001

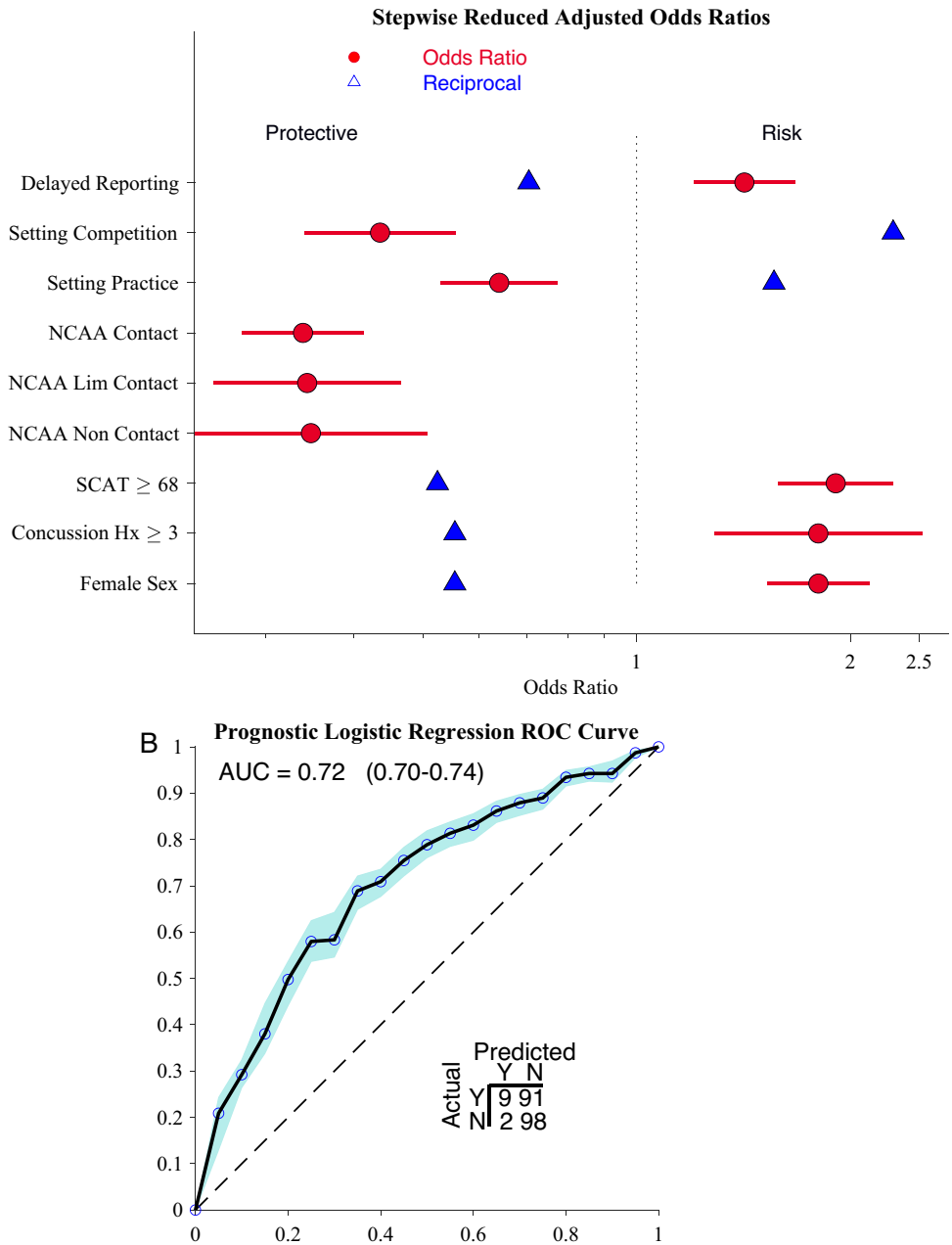

Fig. 3 **A** Adjusted odds ratios for each of the following factors: SCAT symptom score > 45, prior concussion history (≥ 3), sex, noncontact sport, limited contact sport, contact sport, concussion in competition, concussion in practice, and delayed reporting. A forest plot is given with each history's odds ratio and reciprocal. Using PPCS defined as RTP > 29 days **B** Logistic regression model predicting athletes with PPCS. Input variables are prior concussion, high SCAT, female, delayed reporting, practice, competition, contact, noncontact, and limited contact. The AUC is 0.72 (confidence interval = 0.70–0.74) with a sensitivity of 9% and a specificity of 98%. The cyan shaded region is the 95% confidence interval of the ROC curve. National Collegiate Athletics Association (NCAA), limited contact (Lim Contact), Sport Concussion Assessment Tool (SCAT), history (Hx), receiver operator curve (ROC)

Similar results were found in a model run using the time to asymptomatic rather than time to return to play, with results illustrated in Supplementary Table 3 and Supplementary Fig. 7.

4 Discussion

Analyses simultaneously addressed multiple potential risk and protective factors for PPCS using a dataset of ~ 47,000 unique student athletes. Mental health history, high BSI-18 score, high SCAT symptom score, prior concussion history, LOC, female sex, outside of sport concussion, cadet, and delayed reporting are associated with increased time to RTP and increased PPCS risk. In contrast, learning disability (with or without a neurostimulant Rx), migraine history, contact sport, limited contact sport, noncontact sport, and injury setting are associated with decreased time to RTP and lowered PPCS risk. A prognostic stepwise LR had poor sensitivity but high specificity, with an AUC of 0.72.

Our results are consistent with the literature for the following risk factors of slow recovery: prior concussion [4, 7], female sex [9, 13, 18, 19, 28], high SCAT [18, 19], and delayed reporting [15, 22, 23]. They are also consistent with the literature for the following protective factors: learning

disability [3], sport type [4, 13], and injury setting [3]. We found mental health, migraine, and LOC to be marginal factors for recovery (OR = 0.83–1.2). Several previous studies have also found LOC [4, 18, 27, 39] and migraine [10, 40] to have marginal effects. Some notable studies are inconsistent with our results, however, including those reporting that mental health [7, 9, 27], migraine [9, 41], learning disability [4, 9], and LOC [4, 9] are risk factors for PPCS. Additionally, some studies report migraine [4] as protective, while learning disability [7, 9, 27, 42], sport type [19], and concussion history [39] were found to be marginal factors. Our study is the first to identify a high BSI-18 score as a risk factor.

About 4% of our healthy collegiate participants as well as our concussed athletes had been diagnosed with anxiety or depression, and we defined these two groups as positive for mental health history. We did not find an association between mental health history and prolonged recovery or higher symptom severity (Supplementary Fig. 6). This is in contrast to other large cohort studies that found mental health to be associated with worse symptoms, a decline in quality of life, and neuropsychological impairment in pediatric/adolescent populations [43, 44]. It remains to be determined whether this contrast is due to the low prevalence of anxiety and depression in collegiate athletes [45].

A natural hypothesis for most clinicians in sports medicine is that there should be an association between SCAT symptom severity score and RTP. In fact, this has been proposed in the literature [46]. The correlation analysis in Supplementary Fig. 6 does not support this hypothesis, with a Pearson correlation of $r = 0.22$. For this reason, we stratified high symptom severity scores into the top 20th percentile, where we then did find a modest risk factor (OR = 1.91) for those who were markedly symptomatic.

4.1 Prognostic Modeling

An overall goal in concussion/mTBI management is to have prognostic models of outcomes which help both the clinical team and the patient. In SRC, Morgan and colleagues [27] studied 80 recovered SRC and 40 PPCS children and adolescents in a community concussion clinic. Using stepwise LR, they reported that mood disorder and delayed symptoms are the strongest predictors of PPCS with a sensitivity of 55% and specificity of 93%. Chu and colleagues [28] studied 655 concussed children and adolescents with machine learning and reported identifying PPCS with an AUC of 0.81.

In the emergency department (ED), mTBI prognostic literature, and following the work of Silverberg and colleagues [47], Mikolic and colleagues [48] tested three different prognostic models [49–51] using the CENTER-TBI dataset, finding AUCs of 0.75–0.76. Similarly, in the ED and civilian mTBI setting utilizing LR and machine learning models, PPCS was predicted with AUCs in the 0.66–0.80 range, sensitivities between 50% and 60%, and specificities in the 90% range [9, 52–55]. Whether models of ED mTBI can be applied to SRC remains undetermined. As pointed out by Silverberg and colleagues [47], a good SRC model will need to be confirmed in a separate multicenter study. In summary, within the SRC literature, clinical prognostic models, including the model presented here (AUC = 0.72), have poor prognostic capabilities so far. This is in contrast to a SRC neuroimaging prognostic model, which has an AUC of 0.90, sensitivity of 79% and specificity of 100% [56]. A good model would have both sensitivities and specificities in the 85% + range and an AUC above 0.90.

4.2 Limitations

Medical histories for mental health, learning disability, prior concussion, LOC, and migraine were self-reported. Further, mental health was operationalized as a combination of anxiety and depression, and learning disability was a combination of ADD, ADHD, dyslexia, and other learning disabilities. The BSI-18 and SCAT symptom scores are self-reported, subjective symptom checklists. There are missing RTP times because RTP is complicated by factors such as the season ending, medical disqualification, school breaks, graduation, or school transfer. It is also worth noting that, when a season ends, patient contact is less frequent and RTP criteria may change if the athlete has a 1–2 week break before off-season practice begins. Also, no information about an athlete (e.g., sport, sex) is blinded to the sports medicine staff in determining RTP time. The analysis presented here did not include some common concussion assessment tools (e.g., BESS, SAC, and ImPACT) because of their poor effect sizes and AUC [57]. The analysis also did not examine the social construct of race or socioeconomic variables. The CARE Consortium did not measure exposure, so measuring concussion risk per sport or sport category is problematic. Effectively all the measures in this study (e.g., RTP, medical histories, and symptom inventories) are qualitative variables that are vulnerable to patient compliance, patient memory, patient bias, and examiner bias.

Our analysis focused on the RTP time point rather than the asymptomatic time point. Focusing on RTP was done because of the intense and systematic attention that the medical staff puts into the RTP decision at the collegiate sports level. At the collegiate level, clearance for unrestricted RTP is typically a multi-clinician decision after careful observation for several days. This is in contrast to the asymptomatic time point, which effectively becomes a self-reported checkbox that triggers the RTP process. The noncollegiate sport literature typically defines PPCS

according to the asymptomatic time point rather than the RTP time point, and this difference needs to be noted.

Large multicenter studies have numerous limitations, including intersite variability, intrasite variability, and the challenges of large, real-world testing environments. This study focuses on RTP as the outcome measure. Although the CARE study standardized the RTP process on the SCAT3 protocol, the qualitative nature of RTP decisions creates inter- and intrasite differences [29]. Studies on collegiate athletes produce a self-selection bias for very healthy people with few comorbidities. The generalizability of these results to the civilian adult noncollegiate athlete and adult nonathletes is unknown.

5 Conclusions

All the preexisting conditions examined produced small to medium effect sizes on baseline examinations; two (mental health and sex) had small-to-medium effect sizes on post-injury examinations, and three (mental health, learning disability, and sex) had small effect sizes on RTP. As per previous studies, some preexisting conditions increased PPCS risk, while others had equally large protective associations. These were not compatible with many prior studies. The larger odds ratios (e.g., concussion in competition and contact sport) were modest. Protective odds ratios of 0.48 or 0.32 indicate an individual is at 2.1 or 3.1 greater odds of not developing PPCS. Notably, this study supports maintaining an athlete on their learning disability medication post-injury. A multivariate prognostic logistic regression model has a very low sensitivity for predicting PPCS, although it does have a very high specificity for who will not develop PPCS.

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Data Availability All of the analysis code is publicly available at: https://github.com/Nicholas-Port/Persistent-Post-Concussion-Symptoms. Most of the CARE 1.0 dataset is available at: https://fitbir.nih.gov/. The additional CARE 2.0 dataset should be available on the Federal Interagency Traumatic Brain Injury Research (FITBIR) repository in the future.

Author Contributions LTR and NLP made substantial contributions to the conception of the study, the acquisition and interpretation of data, the drafting of the work, critical revision for important intellectual content, and the provision of final approval of the version published. They are accountable for all aspects of the work. GB and FP made substantial contributions to the conception of the study, the analysis and interpretation of data, critical revision for important intellectual content, and the provision of final approval of the version published. PFP, SPB, TMA, and MAMC made substantial contributions to the conception of the study, the acquisition and interpretation of data,

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